

# Atoms & Nuclear Radiation.

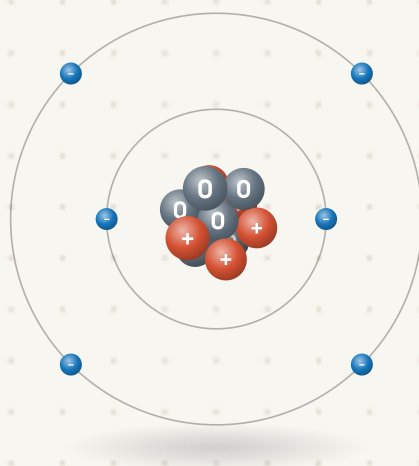
## Lesson 07

## Properties and uses of radiation

A smoke alarm uses alpha radiation, a paper mill uses beta and a hospital uses gamma. Why not use the same radiation for all three?

### IN THIS LESSON YOU WILL

- compare the ionising power, range in air and penetration of alpha, beta and gamma
- explain why more ionising radiation is less penetrating
- identify radiation from absorber data
- choose a suitable source for a use and justify the choice



NAME

CLASS

TEACHER

TARGET GRADE

# How this booklet works.

This is the booklet for **Lesson 7** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.



## DO NOW

Quick questions from earlier lessons. Remembering again is what makes it stick.



## LEARN IT

Explanations, diagrams and worked examples. Read this again when you revise.



## TRY IT

Practice that builds up: first with support, then on your own, then a challenge.



## LAB

Record what you find in a simulation or a model experiment.



## EXAM CORNER

AQA-style questions with a tip on how the marks are given.



## REVISE IT

Summary, cover-and-answer questions and an “I can” check to colour in.



**Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the “Cover the page” questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

## ● The route through the unit (this lesson is highlighted)

01 Inside the atom

03 From plum pudding to the nucleus

05 Radioactive decay

**07 Properties and uses of radiation**  
Ionising power, range and penetration;  
choosing the right source (slides 32–34)

09 Half-life calculations

02 Isotopes and ions

04 Energy levels and the nucleus

06 Nuclear equations

08 Half-life and random decay

10 Contamination and irradiation

Teaching deck: [panphy.app/gcsephy/decks/atoms-and-radiation](https://panphy.app/gcsephy/decks/atoms-and-radiation). Simulations:

[panphy.app/simulations/atomic\\_models.html](https://panphy.app/simulations/atomic_models.html) and [panphy.app/simulations/nuclear\\_decay.html](https://panphy.app/simulations/nuclear_decay.html)

# Properties and uses of radiation.

AQA Physics 4.4.2.1; Combined Science 6.4.2.1

## BIG QUESTION

A smoke alarm uses alpha radiation, a paper mill uses beta and a hospital uses gamma. Why not use the same radiation for all three?

## BY THE END YOU CAN

- compare the ionising power, range in air and penetration of alpha, beta and gamma
- explain why more ionising radiation is less penetrating
- identify radiation from absorber data
- choose a suitable source for a use and justify the choice

ionising

ion

penetration

range

absorbed

smoke alarm

thickness gauge

tracer

## SLIDES AND SIMULATIONS

## Teaching deck, slides 32–34

Comparing alpha, beta and gamma, penetration, and choosing a source.

[panphy.app/gcsephy/decks/atoms-and-radiation/#32](https://panphy.app/gcsephy/decks/atoms-and-radiation/#32)

## Nuclear Decay simulation

[panphy.app/simulations/nuclear\\_decay.htm](https://panphy.app/simulations/nuclear_decay.htm)



## RETRIEVE

## Do now

From earlier lessons. No notes!

1

Complete:  ${}_{84}^{210}\text{Po} \rightarrow {}_{82}^{206}\text{Pb} + ?$


2

What happens to the atomic number in beta decay?


3

What is gamma radiation?



## EXPLAIN

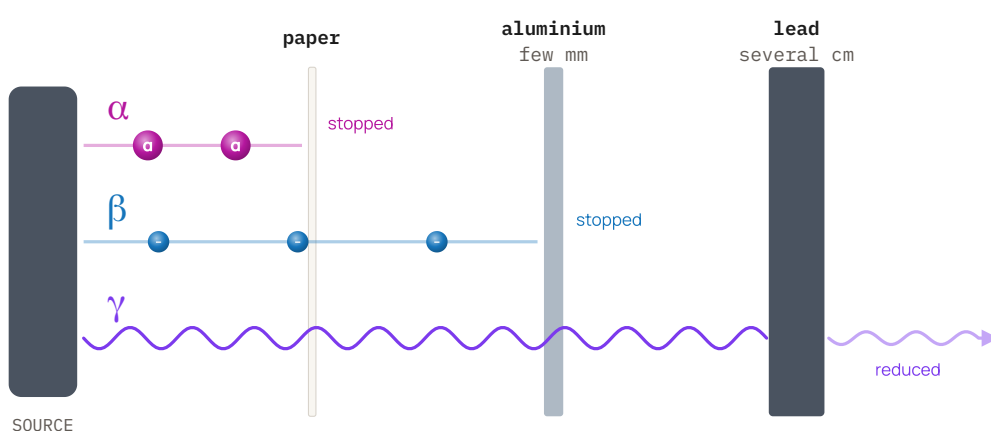
## Ionising and penetrating

### ● Nuclear radiation is ionising

As nuclear radiation passes through a material, it knocks electrons off atoms, turning them into **ions** (Lesson 2). Each ionisation takes some of the radiation's energy. Alpha particles are large

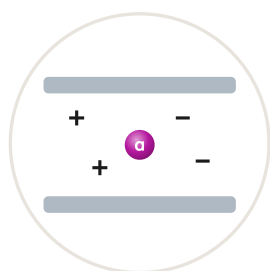
with a +2 charge, so they collide with many atoms and ionise strongly: they lose their energy fast and do not travel far.

|              | Alpha ( $\alpha$ )        | Beta ( $\beta$ )      | Gamma ( $\gamma$ )                |
|--------------|---------------------------|-----------------------|-----------------------------------|
| What is it?  | helium nucleus            | fast electron         | EM wave                           |
| Charge       | +2                        | -1                    | 0                                 |
| Ionising     | strongly                  | moderately            | weakly                            |
| Range in air | a few cm                  | about 1 m             | very far                          |
| Stopped by   | a sheet of paper, or skin | a few mm of aluminium | reduced by thick lead or concrete |



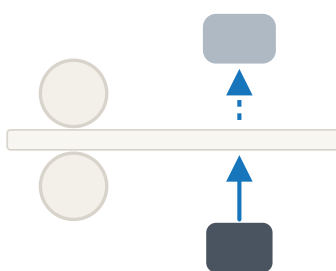
**! More ionising  $\rightarrow$  less penetrating.** Alpha loses its energy fastest, so it is stopped most easily. Gamma ionises weakly, so it travels furthest.

## ● Choosing the right source



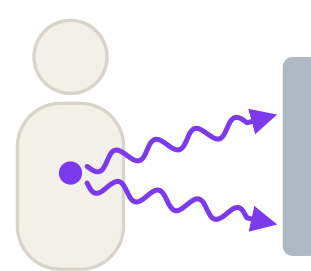
### Alpha: smoke alarm

Alpha ionises the air, so a small current flows. Smoke absorbs the alpha particles, the current falls and the alarm sounds.



### Beta: thickness gauge

Paper absorbs some beta. If the paper gets thicker, the count rate falls, and the rollers adjust.



### Gamma: medical tracer

Gamma passes out of the body to a detector and is only weakly ionising.

To choose a source, think about **penetration** (must it pass through something, or be stopped?), **ionising power**, and **half-life**: how long it keeps giving out radiation (Lesson 8).

## ● Identifying radiation from absorbers

A detector measures the count rate from a source as different absorbers are placed in between. The background count rate is 30 counts per minute.

### WORKED EXAMPLE

Which radiation does this source emit?

| Absorber       | Count rate / counts per minute |
|----------------|--------------------------------|
| none           | 900                            |
| paper          | 460                            |
| 5 mm aluminium | 455                            |
| 2 cm lead      | 90                             |

900 → 460: a big fall, so alpha is present

PAPER

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460 → 455: no real change, so no beta

ALUMINIUM

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455 → 90: falls, but still above background (30), so gamma

LEAD

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ANSWER **alpha and gamma**

### YOUR TURN

Which radiation does this source emit?

| Absorber       | Count rate / counts per minute |
|----------------|--------------------------------|
| none           | 520                            |
| paper          | 515                            |
| 5 mm aluminium | 32                             |
| 2 cm lead      | 31                             |

PAPER

ALUMINIUM

LEAD

ANSWER



PRACTISE

Try it

1

**SUPPORTED** Complete the table.

|                | Alpha    | Beta     | Gamma                   |
|----------------|----------|----------|-------------------------|
| Ionising power |          | moderate |                         |
| Stopped by     |          |          | (reduced by) thick lead |
| Range in air   | a few cm |          |                         |

2

Which radiation is the most strongly ionising? Tick **one** box.

☐ alpha

☐ beta

☐ gamma

3

Draw a line from each use to the radiation it needs.

1 smoke alarm

A gamma

2 paper thickness gauge

B alpha

3 medical tracer

C beta

- 4 Explain why a beta source is used to monitor the thickness of paper, and not alpha or gamma.

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- 5 **ON YOUR OWN** The background count rate is 20 counts per minute. Which radiation does this source emit? Explain using the data.

| Absorber                       | none | paper | 5 mm aluminium | 2 cm lead |
|--------------------------------|------|-------|----------------|-----------|
| Count rate / counts per minute | 640  | 380   | 25             | 21        |

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- 6 Correct Mia's statement.

Mia

Gamma goes through everything, and nothing can stop it.

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- 7 **CHALLENGE** Smoke alarms use americium-241, which has a half-life of 432 years. Suggest why a long half-life is useful here.

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**Get the marks:** Link each property to the reason: ionising → loses energy quickly → short range.

- 8 Explain why alpha radiation is more strongly ionising but less penetrating than gamma radiation. [3 marks]

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- 9 A factory uses a beta source and a detector to control the thickness of aluminium foil. Explain how the count rate shows that the foil has become too thick. [3 marks]

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**SUMMARY**

- Nuclear radiation ionises atoms by knocking off electrons.
- Alpha: strongly ionising, a few cm in air, stopped by paper.
- Beta: moderately ionising, about 1 m in air, stopped by a few mm of aluminium.
- Gamma: weakly ionising, very far in air, reduced by thick lead or concrete.
- More ionising → less penetrating. Choose a source by penetration, ionising power and half-life.

**COVER THE PAGE AND ANSWER**

1 What does ionising mean?

2 What stops beta radiation?

3 Which radiation is used in a smoke alarm?

4 Why is gamma used as a medical tracer?

**I CAN...**

compare the properties of alpha, beta and gamma



explain the link between ionising and penetration



identify radiation from absorber data



choose and justify a source for a use



# Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

## The three particles

Lesson 1

| Particle | Charge | Mass       | Where         |
|----------|--------|------------|---------------|
| Proton   | +1     | 1          | nucleus       |
| Neutron  | 0      | 1          | nucleus       |
| Electron | -1     | very small | energy levels |

## Reading the symbol

Lesson 1

$${}^A_ZX \text{ e.g. } {}^{23}_{11}\text{Na}$$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A - Z; in an atom, electrons = Z
- Atom radius  $\approx 1 \times 10^{-10}$  m; nucleus radius  $< 1/10\,000$  of that

## Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

## Alpha scattering

Lesson 3

| Observation             | Conclusion            |
|-------------------------|-----------------------|
| most straight through   | mostly empty space    |
| some deflected          | positive centre       |
| a very few bounced back | mass in a tiny centre |

## Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

## Nuclear radiation

Lesson 5

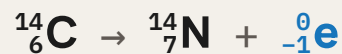
- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- $\alpha$ :  $2p + 2n$ ;  $\beta$ : fast electron, a neutron → proton;  $\gamma$ : EM radiation; n: a neutron

## Nuclear equations

Lesson 6

| Emitted | Symbol              | Mass no.  | Atomic no. |
|---------|---------------------|-----------|------------|
| alpha   | ${}^4_2\text{He}$   | - 4       | - 2        |
| beta    | ${}^0_{-1}\text{e}$ | no change | + 1        |
| gamma   | $\gamma$            | no change | no change  |
| neutron | ${}^1_0\text{n}$    | - 1       | no change  |

Top numbers balance; bottom numbers balance.



## Properties of radiation

Lesson 7

|                     | Alpha       | Beta                  | Gamma                             |
|---------------------|-------------|-----------------------|-----------------------------------|
| <b>Ionising</b>     | strongly    | moderately            | weakly                            |
| <b>Range in air</b> | a few cm    | about 1 m             | very far                          |
| <b>Stopped by</b>   | paper, skin | a few mm of aluminium | reduced by thick lead or concrete |
| <b>Typical use</b>  | smoke alarm | thickness gauge       | medical tracer                    |

More ionising → less penetrating.

### REFERENCE

## Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

**Ionising:** Able to knock electrons off atoms, turning them into ions.

**Penetration:** How far radiation can travel through materials before it is absorbed.

## CHECK YOURSELF

# Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

## DO NOW

1.  ${}^4_2\text{He}$  (an alpha particle)
2. It increases by 1.
3. Electromagnetic radiation from the nucleus.

## COVER THE PAGE

1. Knocking electrons off atoms, making ions.
2. A few mm of aluminium.
3. Alpha.
4. It passes out of the body to a detector and is only weakly ionising.